Channel Coding

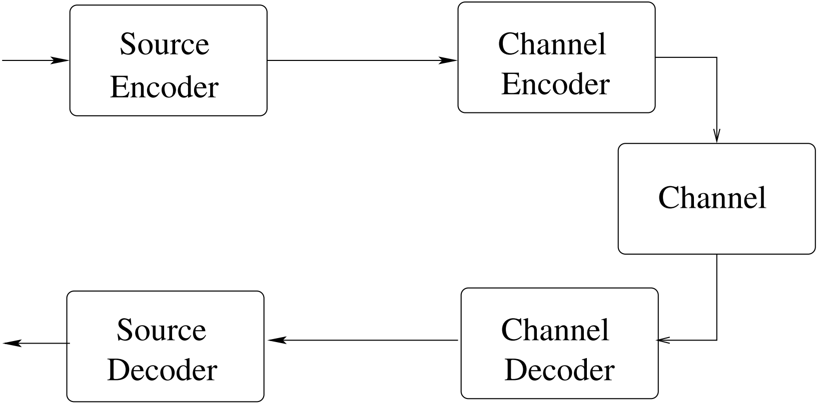
Bikash Kumar Dey[1]

[1] Indian Institute of Technology Bombay, Mumbai, India

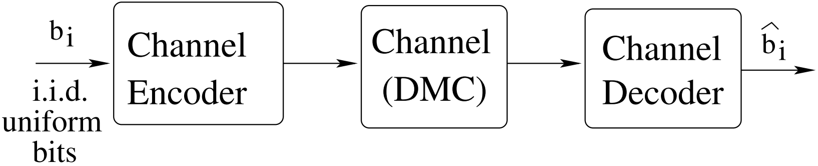
EE 708: Information Theory and Coding

Basics

The Communication System



The channel coding problem



- Input message: i.i.d. uniform (source-channel separation)
- DMC without feedback: to be generalized later

Notation

- x_i : i-th component
- x^i : $(x_1, x_2, \cdots, x_i)$
- Input alphabet $\mathcal{X}$, output alphabet $\mathcal{Y}$: both finite

DMC without feedback

- Memoryless:

$$p(y_k|x^k, y^{k-1}) = p(y_k|x_k)$$

- No feedback:

$$p(x_k, \ldots, x_n|x^{k-1}, y^{k-1}) = p(x_k, \ldots, x_n|x^{k-1})$$

DMC without feedback

$$p(y^n|x^n) = \prod_{i=1}^{n} p(y_k|x^n, y^{k-1})$$

Now

$$\begin{aligned} p(y_k|x^n, y^{k-1}) &= \frac{p(x_{k+1}^n|x^k, y^k)p(y_k|x^k, y^{k-1})p(x^k, y^{k-1})}{p(x_{k+1}^n|x^k, y^{k-1})p(x^k, y^{k-1})} \\ &= \frac{p(x_{k+1}^n|x^k)p(y_k|x_k)}{p(x_{k+1}^n|x^k)} \qquad (1) \\ &= p(y_k|x_k) \end{aligned}$$

In (**??**), numerator 1st term and denominator $\Leftarrow$ no feedback. numerator 2nd term $\Leftarrow$ memoryless. So, we have

$$\boxed{p(y^n|x^n) = \prod_{i=1}^{n} p(y_i|x_i)}$$

Examples of channels

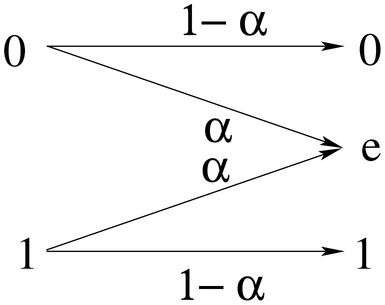
Noiseless channel

$$0 \longrightarrow 0$$

$$1 \longrightarrow 1$$

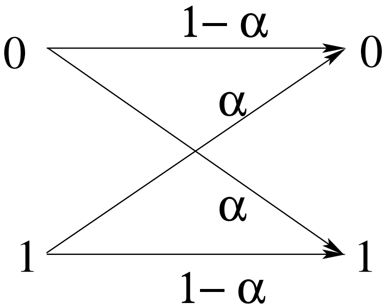
Examples of channels

Erasure channel



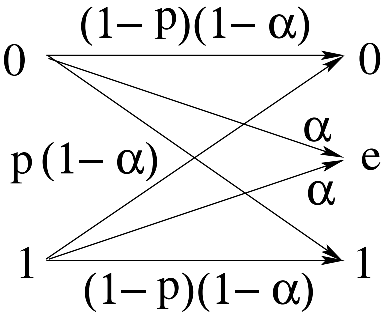
Examples of channels

Binary Symmetric Channel



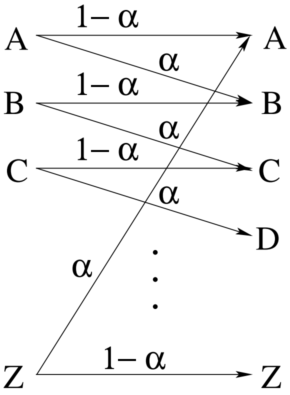
Examples of channels

Binary Error and Erasure Channel

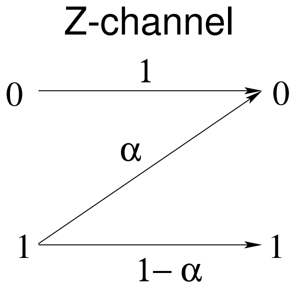


Examples of channels

Noisy Typewriter



Examples of channels



Channel coding theorem

Channel code

Definition

An (M, n) code has

- an index set $\{1, 2, \ldots, M\}$. The message U takes values from this index set uniformly.
- an encoding function $x^n : \{1, 2, \ldots, M\} \rightarrow \mathcal{X}^n$. The codewords are denoted as $x^n(1), x^n(2), \ldots, x^n(M)$. The set of codewords, i.e., the image of the encoding map is called the code or the codebook.
- a decoding function $g : \mathcal{Y}^n \rightarrow \{1, 2, \ldots, M\}$.

Rate of the code: $R = \frac{\log M}{n}$ bits per channel use.

Probability of error

For a given code $\mathcal{C}$, and a message i, the probability of error

$$
\begin{aligned}
P_{e,\mathcal{C},i}^{(n)} &= Pr\{\{y^n : g(y^n) \neq i\} | X^n = x^n(i)\} \\
&= \sum_{y^n : g(y^n) \neq i} p(y^n | x^n(i)).
\end{aligned}
$$

The maximum probability of error:

$$
P_{e,\mathcal{C},\max}^{(n)} = \max_i P_{e,\mathcal{C},i}
$$

The average probability of error for a code:

$$
P_{e,\mathcal{C}}^{(n)} = \frac{1}{M} \sum_i P_{e,\mathcal{C},i}
$$

Achievable rates

Achievable rates

A rate R is *achievable* if there exists a sequence of $(\lceil 2^{nR} \rceil, n)$ codes such that $P_{e,\mathcal{C}}^{(n)} \to 0$ as $n \to \infty$.

Capacity of a channel

Supremum of all achievable rates.

Channel coding theorem

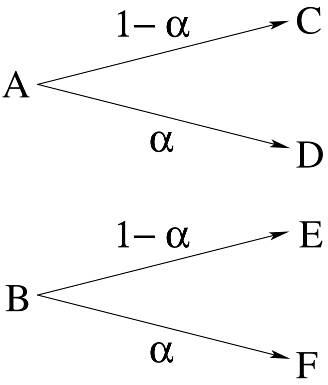
Channel coding theorem

Capacity of a channel is given by

$$C = \max_{p(x)} I(X;Y).$$

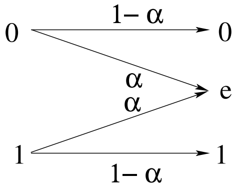
Calculation of capacity: examples

For channels with non-overlapping outputs,
$C = \log |\mathcal{X}|$



Calculation of capacity: BEC

$$
\begin{aligned}
C &= \max_{p(x)} I(X;Y) \\
&= \max_{p(x)} (H(Y) - H(Y|X)) \\
&= \max_{p(x)} H(Y) - H(\alpha)
\end{aligned}
$$



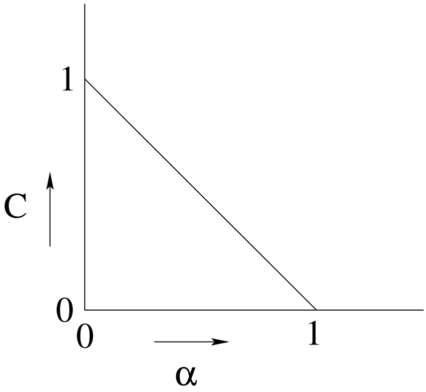
Now, by grouping lemma, if $Pr\{X = 0\} = \pi$, then

$$
\begin{aligned}
H(Y) &= H(Pr\{e\}) + (1 - Pr\{e\})H(Pr\{0\})) \\
&= H(\alpha) + (1 - \alpha)H(\pi)
\end{aligned}
$$

So,

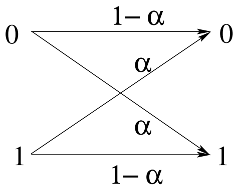
$$
\begin{aligned}
C &= \max_{\pi} (1 - \alpha)H(\pi) \\
&= 1 - \alpha
\end{aligned}
$$

Calculation of capacity: BEC



Calculation of capacity: BSC

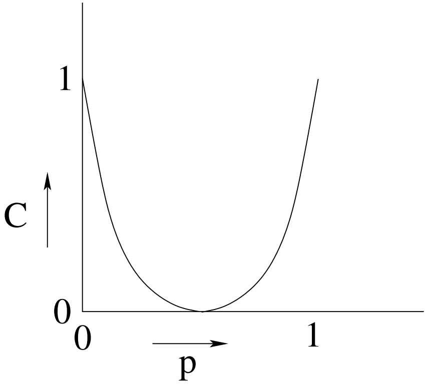
$$\begin{aligned} I(X;Y) &= (H(Y) - H(Y|X)) \\ &= H(Y) - \sum_x p(x) H(Y|X=x) \\ &= H(Y) - \sum_x p(x) H(p) \\ &= H(Y) - H(p) \end{aligned}$$



This is maximized by $Pr\{X=0\} = 1/2$, and

$$C = 1 - H(p)$$

Calculation of capacity: BSC



Symmetric Channel

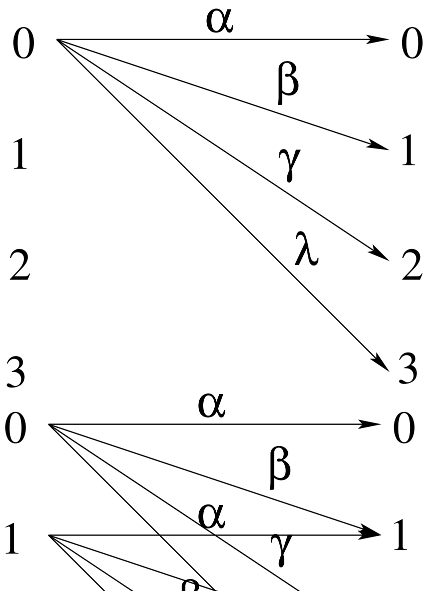
The transition matrix is of the form:

$$p(y|x) = \begin{pmatrix} 0.1 & 0.3 & 0.6 \\ 0.6 & 0.1 & 0.3 \\ 0.3 & 0.6 & 0.1 \end{pmatrix}$$

Definition: Symmetric Channel

Rows are permutations of each other, and columns are permutations of each other.

Symmetric Channel



$$Y = X + Z \bmod 4$$

$$p(y|x) = \begin{pmatrix} \alpha & \beta & \gamma & \lambda \\ \lambda & \alpha & \beta & \gamma \\ \gamma & \lambda & \alpha & \beta \\ \beta & \gamma & \lambda & \alpha \end{pmatrix}$$

Symmetric Channel: Capacity

$$
\begin{aligned}
I(X;Y) &= H(Y) - H(Y|X) \\
&= H(Y) - H(\mathbf{r}) \\
&\leq \log|\mathcal{Y}| - H(\mathbf{r})
\end{aligned}
$$

Now,

$$
\begin{aligned}
p(y) &= \sum_{x\in\mathcal{X}} p(y|x)p(x) \\
&= \frac{1}{|\mathcal{X}|} p(y|x) \\
&= \frac{c}{|\mathcal{X}|} = \frac{1}{|\mathcal{Y}|}
\end{aligned}
$$

if every column sum of $p(y|x)$ is constant c.

Weakly Symmetric Channel

So,

$$C = \log |\mathcal{Y}| - H(\mathbf{r})$$

if it is a *weakly symmetric channel*.

Definition

A channel for which all the rows of the transition matrix are permutations of each other, and the sum of every column is same.

Example: (1) $Y = X + Z \bmod m$ for some m.
(2)

$$p(y|x) = \begin{pmatrix} 1/3 & 1/6 & 1/2 \\ 1/3 & 1/2 & 1/6 \end{pmatrix}$$

Channel coding theorem: achievability proof

Jointly typical sequences

$(X, Y) \sim p(x, y)$.

Definition: jointly typical sequence

A sequence $(x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n$ is said to be (strongly) ϵ-typical if it satisfies

$$\left| T_{x^n, y^n}(x, y) - p(x, y) \right| < \epsilon p(x, y) \qquad \forall (x, y) \in \mathcal{X} \times \mathcal{Y}$$

$\mathcal{T}_\epsilon^{(n)} :=$ the set of jointly typical sequences

Note that if x^n, y^n are jointly typical, then x^n and y^n are also individually typical.

Weakly jointly typical sequences

$(X, Y) \sim p(x, y)$.

Definition: Weakly jointly typical sequence

A sequence $(x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n$ satisfying

$$\left| -\frac{1}{n} \log p(x^n) - H(X) \right| < \epsilon$$
$$\left| -\frac{1}{n} \log p(y^n) - H(Y) \right| < \epsilon$$
$$\left| -\frac{1}{n} \log p(x^n, y^n) - H(X, Y) \right| < \epsilon$$

Properties

If (X^n, Y^n) is drawn i.i.d. from $p(x, y)$, then

- $Pr\{A_\epsilon^{(n)}\} \to 1$ as $n \to \infty$.
- $\left|A_\epsilon^{(n)}\right| \leq 2^{n(H(X,Y)+\epsilon)}$.
- For any $\delta > 0$, $\left|A_\epsilon^{(n)}\right| \geq (1-\delta)2^{n(H(X,Y)-\epsilon)}$ for large enough n.
- If $(\tilde{X}^n, \tilde{Y}^n) \sim p(x^n)p(y^n)$, then

  $$Pr\left((\tilde{X}^n, \tilde{Y}^n) \in A_\epsilon^{(n)}\right) \leq 2^{-n(I(X;Y)-3\epsilon)}$$

  Also, for any $\delta > 0$ and for sufficiently large n,

  $$Pr\left((\tilde{X}^n, \tilde{Y}^n) \in A_\epsilon^{(n)}\right) \geq (1-\epsilon)2^{-n(I(X;Y)+3\epsilon)}$$

Properties: Proof

(1),(2) and (3) follow similar to the single variable case.

$$
\begin{aligned}
(4) Pr\left((\tilde{X}^n, \tilde{Y}^n) \in A_\epsilon^{(n)}\right) &= \sum_{(x^n, y^n) \in A_\epsilon^{(n)}} p(x^n) p(y^n) \\
&\leq 2^{n(H(X,Y)+\epsilon)} 2^{-n(H(X)-\epsilon)} 2^{-n(H(Y)-\epsilon)} \\
&\leq 2^{-n(I(X;Y)-3\epsilon)}
\end{aligned}
$$

For large enough n,

$$
\begin{aligned}
Pr\left((\tilde{X}^n, \tilde{Y}^n) \in A_\epsilon^{(n)}\right) &= \sum_{(x^n, y^n) \in A_\epsilon^{(n)}} p(x^n) p(y^n) \\
&\geq (1-\delta) 2^{n(H(X,Y)-\epsilon)} 2^{-n(H(X)+\epsilon)} 2^{-n(H(Y)+\epsilon)} \\
&\leq (1-\delta) 2^{-n(I(X;Y)+3\epsilon)}
\end{aligned}
$$

More properties: Proofs left as exercise

Define

$$\mathcal{T}_\epsilon^{(n)}(Y|x^n) := \{y^n|(x^n, y^n) \in \mathcal{T}_\epsilon^{(n)}(X, Y)\}$$

1. For every $x^n \in \mathcal{X}^n$,

$$|\mathcal{T}_\epsilon^{(n)}(Y|x^n)| \leq 2^{n(H(Y|X)+\delta(\epsilon))},$$

   where $\delta(\epsilon) = \epsilon H(Y|X)$.

2. Let $X \sim p(X)$ and $Y = g(X)$. Let $x^n \in \mathcal{T}_\epsilon^{(n)}(X)$. Then $y^n \in \mathcal{T}_\epsilon^{(n)}(Y|x^n)$ if and only if $y_i = g(x_i)$ for all $i \in [1 : n]$.

More properties

Conditional typicality lemma

Let $(X, Y) \sim p(x, y)$. Suppose that $x^n \in \mathcal{T}_{\epsilon'}^{(n)}(X)$ and $Y^n \sim p(y^n|x^n) = \prod_i p_{Y|X}(y_i|x_i)$. Then, for every $\epsilon > \epsilon'$,

$$\lim_{n\to\infty} P\{(x^n, Y^n) \in \mathcal{T}_\epsilon^{(n)}(X, Y)\} = 1$$

I

f $x^n \in \mathcal{X}^n$ and $\epsilon' < \epsilon$, then for sufficiently large n,

$$|\mathcal{T}_\epsilon^{(n)}(Y|x^n)| \geq (1-\epsilon)2^{n(H(Y|X)-\delta(\epsilon))},$$

where $\delta(\epsilon) = \epsilon H(Y|X)$.

Channel coding theorem: achievability proof

- Choose any rate $R < C$. There is a $p(x)$ such that $I(X;Y) > R$.
- Choose a $(\lceil 2^{nR} \rceil, n)$ code by picking all the $n.\lceil 2^{nR} \rceil$ code components i.i.d. $\sim p(x)$. Denote the codewords as $X^n(1), X^n(2), \ldots, X^n(\lceil 2^{nR} \rceil)$.
- The code is known to the encoder and decoder.
- The message $W \in \{1, 2, \ldots, \lceil 2^{nR} \rceil\}$ is encoded by the codeword $X^n(W)$ and the codeword is transmitted through the channel.
- *Typicality Decoding*: After receiving Y^n, the decoder chooses a codeword which is jointly typical with Y^n. If there are more than one jointly typical codeword, or none, then the decoder declares an error.

Error analysis

Prob. of error, averaged over messages and codes:

$$\begin{aligned} P_e^n &\triangleq \sum_{\mathcal{C}_n} \sum_{w} Pr(w, \mathcal{C}_n) P_{e,\mathcal{C}_n,w}^{(n)} \\ &= \sum_{\mathcal{C}} Pr\{\mathcal{C}_n\} P_{e,\mathcal{C}_n}^{(n)} \end{aligned}$$

Clearly, $\exists \mathcal{C}_n$ s.t. $P_{e,\mathcal{C}_n}^{(n)} \leq P_e^n$. We will show that $P_e^n \to 0$ as $n \to 0$.

$\Rightarrow$

There exists a sequence of codes $\mathcal{C}_n$ for which $P_{e,\mathcal{C}_n}^{(n)} \to 0$ as $n \to 0$.

Error analysis

$$
\begin{aligned}
P_e^n &\triangleq \sum_{\mathcal{C}_n} \sum_{w} Pr(w, \mathcal{C}_n) P_{e,\mathcal{C}_n,w}^{(n)} \\
&= \sum_{w} Pr\{W = w\} P_{e,w}
\end{aligned}
$$

Note that $P_{e,w}$ is same for all w for random code construction. So,

$$
P_e^n = P_{e,1}
$$

Evaluating $P_{e,1}$: $X^n(1)$ is transmitted

When does an error happen?

- The transmitted codeword $X^n(1)$ is not jointly typical with the received vector.
- Another codeword $X^n(i)$, $i \neq 1$ is jointly typical with the received vector.

Define the event:

$$E_i \stackrel{\triangle}{=} \{(X^n(i), Y^n) \in A_\epsilon^{(n)}\}$$

Then

$$\begin{aligned} P_{e,1} &= Pr\{E_1^c \cup E_2 \cup E_3 \cup \cdots \cup E_{\lceil 2^{nR} \rceil} | W = 1\} \\ &\leq Pr\{E_1^c | W = 1\} + \sum_{i=2}^{\lceil 2^{nR} \rceil} Pr\{E_i | W = 1\} \end{aligned}$$

Evaluating $P_{e,1}$: $X^n(1)$ is transmitted

- $X^n(i)$ are independently chosen with components i.i.d. from $p(x)$,
- $(X^n(1), Y^n)$ is a i.i.d. sequence from $p(x, y)$

So by AEP,

$$Pr\{E_1^c | W = 1\} \to 0 \text{ as } n \to \infty$$

- $(X^n(i), Y^n)$ is i.i.d. from $p(x)p(y)$.

So,

$$Pr\{E_i | W = 1\} \leq 2^{-n(I(X;Y)-3\epsilon)}$$

Evaluating $P_{e,1}$: $X^n(1)$ is transmitted

For large enough n,

$$
\begin{aligned}
P_{e,1} &\leq Pr\{E_1^c | W = 1\} + \sum_{i=2}^{\lceil 2^{nR} \rceil} Pr\{E_i | W = 1\} \\
&\leq \epsilon + \sum_{i=2}^{\lceil 2^{nR} \rceil} 2^{-n(I(X;Y)-3\epsilon)} \\
&\leq \epsilon + (\lceil 2^{nR} \rceil - 1) 2^{-n(I(X;Y)-3\epsilon)} \\
&\leq \epsilon + 2^{nR}.2^{-n(I(X;Y)-3\epsilon)} \\
&\leq \epsilon + 2^{-n\epsilon}
\end{aligned}
$$

if $\epsilon < \frac{I(X;Y)-R}{4}$. So, for large enough n,

$$P_{e,1} \leq 2\epsilon$$

Error analysis

We have proved

1. P_e^n can be made arbitrarily small
2. $\Rightarrow$ There exists a sequence of codes $\mathcal{C}_n$ of any fixed rate $R < I(X;Y)$ of increasing length so that $P_{e,\mathcal{C}_n} \to 0$ as $n \to \infty$. $\boxed{\text{Achievability of the channel coding theorem}}$
3. By choosing $p(x) = p^*(x)$, one can transmit at any rate $R < C$ with $P_{e,\mathcal{C}_n} \to 0$ as $n \to \infty$.

Achievability under maximum error probability

Achievability under maximum error probability

1. By throwing away ("expurgating") worst half of the codewords, one gets a code with rate $R - \frac{1}{n} \to R$ where each codeword has $P_{e,\mathcal{C}_n,i} < 4\epsilon$. So,
2. There exists a sequence of codes $\mathcal{C}_n$ of rate $R_n \to R < C$ of increasing length so that the maximum error probability $P_{e,\mathcal{C}_n,m} \to 0$ as $n \to \infty$.

Proof of the converse

Fano's Inequality

$$H(W|\hat{W}) \leq H(P_e) + P_e log(2^{nR})$$

Lemma

For any $p(x^n), I(X^n; Y^n) \leq nC$.

Proof of the lemma

First recall that we proved for memoryless and feedback-free channels,

$$p(y_i|y_1,\dots,y_{i-1},x^n) \quad = p(y_i|x_i)$$

i.e., $X^n, Y^{i-1} \leftrightarrow X_i \leftrightarrow Y_i$ forms a Markov chain.
Thus,

$$H(Y_i|Y_1,\dots,Y_{i-1},X^n) \quad = H(Y_i|X_i)$$

Proof of the lemma

$$
\begin{aligned}
I(X^n; Y^n) &= H(Y^n) - H(Y^n|X^n) \\
&= H(Y^n) - \sum_{i=1}^{n} H(Y_i|Y_1, \ldots, Y_{i-1}, X^n) \\
&= H(Y^n) - \sum_{i=1}^{n} H(Y_i|X_i) \ (\Leftarrow \text{memoryless,no feedback}) \\
&\leq \sum_{i=1}^{n} H(Y_i) - \sum_{i=1}^{n} H(Y_i|X_i) \\
&= \sum_{i=1}^{n} I(X_i; Y_i) \\
&\leq nC.
\end{aligned}
$$

Proof of the converse

$$\begin{aligned}
nR &= H(W) \\
&= H(W|\hat{W}) + I(W;\hat{W}) \\
&\leq H(P_e) + P_e nR + I(W;\hat{W}) \\
&\leq H(P_e) + P_e nR + I(X^n;Y^n) \\
&\leq H(P_e) + P_e nR + nC \\
\Rightarrow R &\leq P_e R + \frac{H(P_e)}{n} + C
\end{aligned}$$

If for a sequence of codes of rate R, $P_e \to 0$, then we have $R \leq C$.

DMC with feedback

DMC with Feedback

- In a code for channel with feedback, X_i may depend on W, Y^{i-1}.
- Clearly, the feedback capacity $C_{FB} \geq C$.
- The converse needs a proof.

$W \rightarrow Y^n \rightarrow \hat{W}$ - Markov chain

So,

$$I(W; \hat{W}) \leq I(W; Y^n)$$

.

Proof of the converse

A feedback code induces a joint pmf of the form:

$$p(w, x^n, y^n) \quad = 2^{-nR} \prod_{j=1}^{n} p(x_j | w, y^{j-1}) p_{Y|X}(y_j | x_j)$$

So, the joint distribution of

$$\begin{aligned} p(w, x^i, y^i) \quad &= 2^{-nR} \prod_{j=1}^{i} p(x_j | w, y^{j-1}) p_{Y|X}(y_j | x_j) \\ &= p(w, x^i, y^{i-1}) p(y_i | x_i) \end{aligned}$$

where

$$p(w, x^i, y^{i-1}) \quad = \left(2^{-nR} \prod_{j=1}^{i-1} p(x_j | w, y^{j-1}) p_{Y|X}(y_j | x_j) \right) \cdot p(x_i | w, y^{i-1})$$

Proof of the converse

Hence $W, X^{i-1}, Y^{i-1} \leftrightarrow X_i \leftrightarrow Y_i$ forms a Markov chain.

Thus

$$H(Y_i|Y_1, ..., Y_{i-1}, W, X_i) = H(Y_i|X_i)$$

Proof of the converse

$$
\begin{aligned}
I(W;Y^n) &= H(Y^n) - H(Y^n|W) \\
&= H(Y^n) - \sum_{i=1}^{n} H(Y_i|Y_1,\ldots,Y_{i-1},W) \\
&= H(Y^n) - \sum_{i=1}^{n} H(Y_i|Y_1,\ldots,Y_{i-1},W,X_i) \qquad (2) \\
&= H(Y^n) - \sum_{i=1}^{n} H(Y_i|X_i) \qquad \text{(Memoryless Channel)} \\
&\le \sum_{i=1}^{n} H(Y_i) - \sum_{i=1}^{n} H(Y_i|X_i) \\
&= \sum_{i=1}^{n} I(X_i;Y_i) \le nC
\end{aligned}
$$

Eq. (1) follows since X_i is a function of Y^{i-1}, W.

Proof of the converse

So,

$$\begin{aligned} H(W) &\leq H(P_e) + P_e nR + I(W;\hat{W}) \\ \Rightarrow nR &\leq H(P_e) + P_e nR + nC \\ \Rightarrow R &\leq \frac{H(P_e)}{n} + P_e R + C \end{aligned} \tag{3}$$

So, if $P_e \to 0$, then $R \leq C$.
So, $\boxed{C_{fb} = C}$

Source-Channel Separation Theorem

Entropy rate of a source

Let $\mathcal{U} = (U_i)_{i=1}^{\infty}$ be a stationary ergodic source process.
Its entropy rate is defined as:

$$H(\mathcal{U}) := \lim_{L\to\infty} \frac{1}{L} H(U_1, \cdots, U_L) \tag{4}$$

This can be shown to be the same as (See Cover and Thomas, 2nd ed, Ch. 4)

$$H(\mathcal{U}) = \lim_{L\to\infty} H(U_L | U_1, \cdots, U_{L-1}) \tag{5}$$

Note that

$$\begin{aligned} H(U_L | U_1, \cdots, U_{L-1}) &\le H(U_L | U_2, \cdots, U_{L-1}) \\ &= H(U_{L-1} | U_1, \cdots, U_{L-2}) \quad \text{(Stationarity)} \end{aligned}$$

Entropy rate of a source

So,

$$
\begin{aligned}
& H(U_L|U_1, \cdots, U_{L-1}) \text{ is non-increasing} \\
\Rightarrow \quad & \frac{1}{L}H(U^L) = \frac{1}{L}\sum_{l=1}^{L} H(U_l|U_1, \cdots, U_{l-1}) \text{ is non-increasing}
\end{aligned}
$$

So

$$
H(\mathcal{U}) \leq \frac{1}{L}H(U^L) \text{ for any } L \tag{6}
$$

A source can be compressed (using LZ algorithm, for example) upto a rate of $H(\mathcal{U})$.

Separation of source coding and channel coding

Any coding scheme:

$$U^L \xrightarrow{Enc} X^n \xrightarrow{Channel} Y^n \xrightarrow{Dec} V^L$$

-forms a Markov chain.

$$\text{Rate } \alpha = \frac{L}{n}$$

DMC satisfies

$$I(X^n; Y^n) \leq nC = n \cdot \max_{p(x)} I(X; Y)$$

Separation of source coding and channel coding

Separation based scheme:
A source can be transmitted using separate source and channel coding at a rate α symbols/channel use if

$$\alpha = \frac{L}{n} < \frac{C}{H(\mathcal{U})}.$$

Joint source-channel coding: We now argue that no higher rate can be achieved using joint coding.

Source-channel separation theorem

If a rate of α symbols per channel use can be achieved for transmitting a discrete stationary ergodic source over a discrete memoryless channel, then the same rate can be achieved by separate source and channel coding.

Separation of source coding and channel coding

Converse under joint source-channel coding:

We have, from Fano's inequality

$$
\begin{aligned}
& P_e^{(n)} \log_2(|\mathcal{U}|^L - 1) + H(P_e^{(n)}) \geq H(U^L) - I(U^L; V^L) \\
\Rightarrow \quad & L \cdot P_e^{(n)} \log_2 |\mathcal{U}| + H(P_e^{(n)}) \geq H(U^L) - I(X^n; Y^n) \\
\Rightarrow \quad & P_e^{(n)} \log_2 |\mathcal{U}| + \frac{1}{L} H(P_e^{(n)}) \geq \frac{1}{L} H(U^L) - \frac{nC}{L} \\
\Rightarrow \quad & \frac{1}{L} H(U^L) - \frac{nC}{L} \leq 0 \quad \text{(by taking } n, L \to \infty, P_e^{(n)} \to 0\text{)} \\
\Rightarrow \quad & H(\mathcal{U}) \leq \frac{nC}{L} \quad \left(H(\mathcal{U}) \leq \frac{1}{L} H(U^L)\right) \\
\Rightarrow \quad & \alpha = \frac{L}{n} \leq \frac{C}{H(\mathcal{U})}
\end{aligned}
$$

Thank you